12 (a) Write $\frac{4x^2 - 9}{6x + 9} \times \frac{2x}{x^2 - 3x}$ in the form $\frac{ax + b}{cx + d}$ where a, b, c and d are integers.

	$2x$	3
$2x$	$4x^2$	$6x$
-3	$-6x$	-9

$4x^2 - 9 = (2x - 3)(2x + 3)$

1 mark
Factorise the numerator

HCF	$2x$	3
3	$6x$	9

HCF	x	-3
x	x^2	$-3x$

1 mark
Factorise the denominators

$$\frac{(2x - 3)(2x + 3) \times 2x}{3(2x + 3) \times x(x - 3)} = \frac{2(2x - 3)}{3(x - 3)}$$

Cancel x and $(2x + 3)$; $x \neq 0, 3, -1.5$.

1 mark
Cancel factors and expand

$\frac{4x - 6}{3x - 9}$

(3)

(b) Express $\frac{3}{x+1} + \frac{1}{x-2} - \frac{4}{x}$ as a single fraction in its simplest form.

Common denominator: $x(x + 1)(x - 2)$

1 mark
Make all three denominators equal

$$\frac{3x(x - 2) + x(x + 1) - 4(x + 1)(x - 2)}{x(x + 1)(x - 2)}$$

1 mark
Combine the equivalent numerators

	x	-2
x	x^2	$-2x$
1	x	-2

Numerator $= 3x^2 - 6x + x^2 + x$

$- 4(x^2 - x - 2)$

$= 3x^2 - 6x + x^2 + x - 4x^2 + 4x + 8$

$= 8 - x$

$$\frac{8 - x}{x(x + 1)(x - 2)}$$

1 mark
Simplify the numerator fully

$x \neq -1, 0, 2$

(3)

(Total for Question 12 is 6 marks)